

Renewable Energy Zoning Dashboard

User Manual

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1. Introduction

1.1 Purpose and Scope

The Renewable Energy Zoning Dashboard is a GIS-based web application designed for identifying, evaluating, and ranking optimal sites for renewable energy projects. The platform supports three distinct project modes:

- **Solar PV Zoning** — Identifies and scores land parcels suitable for photovoltaic solar installations based on solar irradiation, terrain, land use, and infrastructure proximity.
- **On-Shore Wind Zoning** — Evaluates inland areas for wind turbine deployment using wind resource data, terrain constraints, environmental exclusions, and grid connectivity.
- **Off-Shore Wind Zoning** — Analyzes maritime Exclusive Economic Zones (EEZ) for offshore wind farms, incorporating bathymetry, seabed substrate, marine constraints, port proximity, and subsea infrastructure.

The application guides the user through a structured four-step analytical pipeline — from defining the study area and generating a spatial grid, through multi-layer raster and vector analysis, weighted scoring with hard exclusion constraints, spatial clustering, and finally financial viability assessment. Each step builds upon the results of the previous one, creating a comprehensive site suitability analysis.

1.2 Technology Stack

The dashboard is built on a modern full-stack architecture orchestrated with Docker Compose:

Component	Technology	Description
Frontend	Next.js 14, React 18, Tailwind CSS, Leaflet	Interactive UI with map visualization at port 3000
Backend	Django 4.2, Django REST Framework, Gunicorn	REST API serving analysis endpoints at port 8000
GIS Engines	GeoPandas, Rasterio, GDAL 3.11, NetworkX, Shapely	Spatial analysis, raster/vector processing, graph-based clustering
Deployment	Docker Compose	Two-container setup with shared data volumes

1.3 Four-Step Analytical Pipeline

The entire workflow is organized into four sequential tabs in the dashboard interface, each corresponding to one pipeline step:

Step	Dashboard Tab	Purpose	Key Output
1	 1. Gridization	Define the study area boundary and divide it into uniform rectangular grid cells	Grid of cells covering the selected region

2	 2. Layer Calculation	Analyze each grid cell against multiple GIS raster and vector layers (distances, coverage, statistics)	Per-cell metrics for each configured layer
3	 3. Scoring	Assign weighted scores to each cell; apply hard exclusion constraints; kV layer weights feed into Step 4 connection scoring	FINAL_GRID_SCORE per cell (0–100)
4	 4. Cluster & Aggregation	Group adjacent high-scoring cells into project clusters; score transmission connections; calculate CAPEX, LCOE, and payback period	Ranked clusters with financial feasibility metrics

Results are visualized on an interactive Leaflet map at each stage, and can be downloaded as CSV files for further analysis in external tools such as QGIS or Excel.

1.4 Session Management

When a project mode is selected, the backend assigns a unique Session ID (UUID) stored in the browser's localStorage as dashboard_session_id. This identifier is attached to every subsequent API request via the X-Session-ID HTTP header. All intermediate results — grid data, analysis outputs, scoring tables — are persisted server-side as serialized files (pickle format) under a session-specific directory (temp/sessions/<uuid>/). Session metadata is stored as session_meta.json alongside the pickle files.

Note: Sessions are temporary. If the backend container is restarted or the session is reset, all intermediate results will be lost. Always download important results before closing the application.

1.5 Asynchronous Operations

The three computation-intensive steps — raster/vector analysis (Step 2), scoring (Step 3), and cluster analysis (Step 4) — run as asynchronous background tasks. When triggered, the backend immediately returns a task_id. The frontend continuously polls GET /api/task/<task_id>/progress/ to retrieve the current progress (0–100%), status message, and completion state. A ProcessingOverlay component displays an animated spinner and the latest progress message during execution. This design allows large analyses to run without blocking the browser.

1.6 Getting Started

To launch the dashboard, ensure Docker is installed and run the following command from the project root directory:

```
docker compose up
```

Once the containers are running:

1. Open a web browser and navigate to <http://localhost:3000>
2. The Landing Page will appear with three project mode cards

3. Click on a project mode to begin (e.g., "Select Solar PV Zoning")
4. You will be redirected to the main Dashboard view with four analysis tabs

2. Solar PV Zoning Study — Detailed Software Functionalities

This section provides a comprehensive walkthrough of the Solar PV Zoning mode, covering every screen, button, input field, and interactive element from the moment the application is opened. While this guide focuses on the Solar PV workflow, the On-Shore and Off-Shore Wind modes follow the same four-tab structure with project-specific layers and parameters noted where applicable.

2.0 Home Page — Project Mode Selection

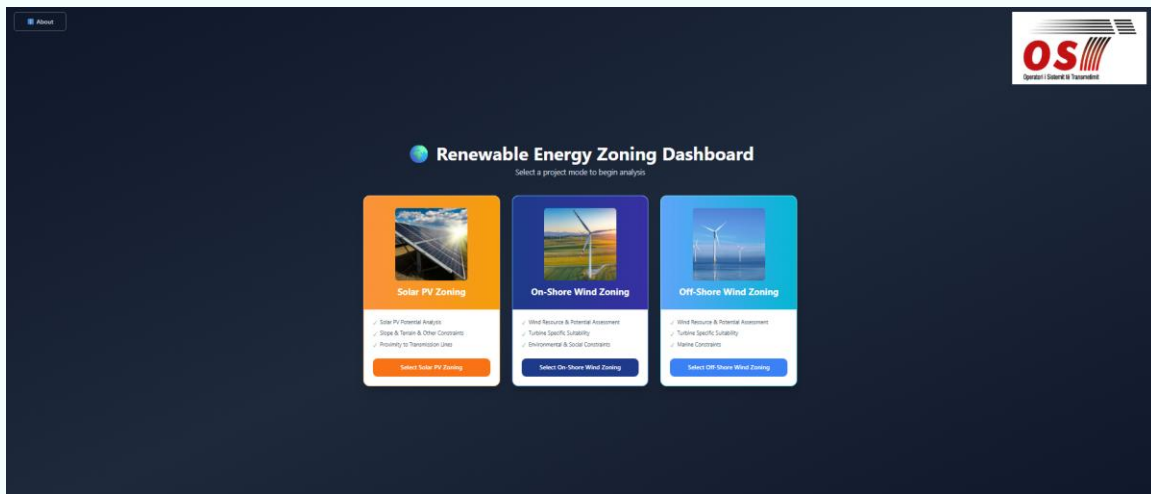


Figure 1: Homepage

Upon navigating to the application URL, the user is presented with the Landing Page. This page serves as the entry point and project mode selector.

2.0.1 Page Layout

The Landing Page features a dark gradient background (slate tones) with the following elements:

- **Main Title:** "Renewable Energy Zoning Dashboard" — displayed prominently at the top center.
- **Subtitle:** "Select a project mode to begin analysis" — guides the user to choose a mode.
- **About Button:** Located in the top-left corner. Labeled "About" — opens an informational modal describing the platform's purpose, pipeline steps, and technical architecture.
- **OST Logo:** Displayed in the top-right corner of the page.
- **Project Mode Cards:** Three large selection cards arranged horizontally (responsive layout).

2.0.2 Project Mode Cards

Each card contains:

- A project-specific icon image (Solar panel, wind turbine on land, or offshore turbine)
- The project mode title (e.g., "Solar PV Zoning")
- A brief list of key analysis capabilities
- A selection button (e.g., "Select Solar PV Zoning")

Card	Title	Key Capabilities	Button Color
Solar PV	Solar PV Zoning	Solar PV Potential Analysis; Slope & Terrain & Constraints; Proximity to Transmission Lines	Orange
On-Shore Wind	On-Shore Wind Zoning	Wind Resource & Potential; Turbine Specific Suitability; Environmental & Social Constraints	Dark Blue
Off-Shore Wind	Off-Shore Wind Zoning	Wind Resource & Potential; Turbine Specific Suitability; Marine Constraints & Seabed Analysis	Cyan/Blue

Clicking a card initiates a backend call (POST /api/project/select/) to create a new session with the selected project type. A loading spinner appears on the card during this process. Upon success, the user is automatically redirected to the main Dashboard.

2.0.3 About Modal

The About modal provides detailed information organized into sections: Overview, How It Works (4-Step Pipeline), Solar PV Mode, On-Shore Wind Mode, Off-Shore Wind Mode, Financial Analysis, Technical Architecture, and Data & Outputs. The modal can be closed by clicking the X button or clicking outside the modal area.

2.0.4 Main Dashboard Layout

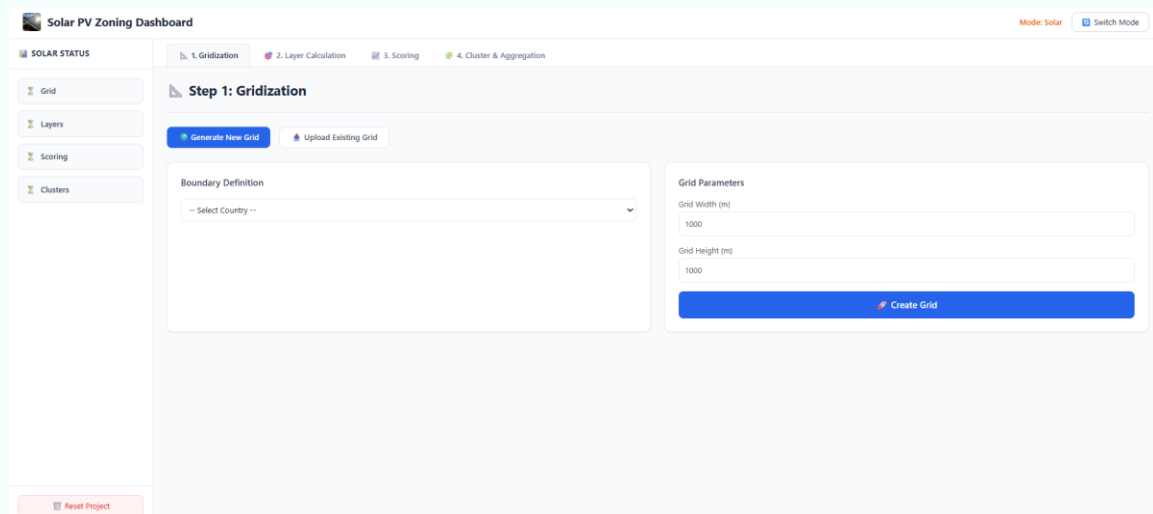


Figure 2: Dashboard overview with sidebar and tab bar visible

After selecting Solar PV mode, the Dashboard page loads. The interface consists of:

Header Bar

- **Color Accent Bar:** A thin colored line at the very top of the page (orange for Solar PV, dark blue for On-Shore Wind, cyan for Off-Shore Wind) provides instant visual identification of the active project mode.
- **Project Icon:** The project-specific icon is displayed on the left side of the header.
- **Application Title:** Shown next to the icon, pulled from the project configuration.
- **Mode Badge:** Displays the active project type (e.g., "Mode: Solar") in the header's right area.
- **Switch Mode Button:** Labeled "Switch Mode" — prompts for confirmation, resets the current session entirely, clears the localStorage session ID, and returns to the Landing Page for a new project selection.

Sidebar (Left Panel)

A fixed-width sidebar (256 pixels) on the left side displays the project status at a glance:

- **Status Title:** Shows the project type followed by "Status" (e.g., "Solar Status").
- **Step Indicators:** Four status rows — Grid, Layers, Scoring, Clusters — each showing a checkmark (✅ completed) or hourglass (⌚ pending) icon. When a step is completed, it also displays the count of items (e.g., "1,250 cells" for Grid, "5 layer(s)" for Layers, cluster count for Clusters).
- **Reset Project Button:** A red button labeled "Reset Project" at the bottom of the sidebar. Prompts for confirmation before clearing all project data while keeping the same project mode active (calls POST /api/project/reset/ with keep_project_type: true).

Tab Bar (Main Content Area)

The main content area occupies the remaining space to the right of the sidebar. At the top, a horizontal tab bar provides navigation between the four pipeline steps:

Tab #	Label	Pipeline Step
1	 1. Gridization	Define study area & create grid
2	 2. Layer Calculation	Raster and vector analysis across all layers
3	 3. Scoring	Weighted scoring, exclusion constraints, kV connection weights
4	 4. Cluster & Aggregation	Clustering, connection scoring, financial analysis

The active tab is highlighted with a distinct background color and bottom border accent. Inactive tabs are shown in muted gray tones and become highlighted on hover.

2.1 Tab 1 — Gridization

1. Gridization2. Layer Calculation3. Scoring4. Cluster & Aggregation

Step 1: Gridization

Generate New GridUpload Existing Grid

Boundary Definition

-- Select Country --

Grid Parameters

Grid Width (m)
1000

Grid Height (m)
1000

Create Grid

Figure 3: Gridization tab — Generate New Grid mode

The Gridization tab is the first step in the analytical pipeline. Its purpose is to define the geographic study area and subdivide it into a uniform grid of rectangular cells. Each cell will later be individually analyzed, scored, and potentially clustered.

2.1.1 Mode Selection

At the top of the tab, two toggle buttons allow the user to choose between:

-  **Generate New Grid:** Create a fresh grid by selecting a geographic boundary and specifying cell dimensions. This is the primary workflow.
-  **Upload Existing Grid:** Import a previously created grid as a CSV file. Useful for re-running analysis on a previously defined study area.

2.1.2 Generate New Grid

When "Generate New Grid" is selected, the interface splits into two columns:

Left Column: Boundary Definition

This section determines the geographic area that will be divided into grid cells.

For Solar PV and On-Shore Wind modes:

- **Select Country:** Choose from a dropdown of European countries derived from the NUTS (Nomenclature of Territorial Units for Statistics) dataset at the national level (LEVL_CODE = 0). The dropdown is populated automatically from the GET /api/countries/ endpoint.
- **Albania Sub-National Drill-Down:** When Albania is selected from the country dropdown, two additional dropdowns appear: (1) Region — populated from GET /api/albania/regions/ listing all ADM1 administrative regions; (2) District — populated from GET /api/albania/districts/?region=... filtered by the selected region, listing ADM2 districts. Selecting a district uses the precise district boundary rather than the entire country boundary for grid generation.

For Off-Shore Wind mode, the boundary source is fixed to EEZ (Exclusive Economic Zone) selection. A dropdown is populated with available European maritime zones from GET /api/eez-zones/.

Note: Once a boundary is selected, an interactive map preview appears below the form, showing the boundary polygon and a real-time grid overlay.

Right Column: Grid Parameters

The grid dimensions depend on the project type:

Solar PV:

Parameter	Input Type	Range	Step	Default	Description
Grid Width (m)	Number	100 – 10,000 m	100 m	1,000 m	Horizontal dimension of each grid cell in meters
Grid Height (m)	Number	100 – 10,000 m	100 m	1,000 m	Vertical dimension of each grid cell in meters

On-Shore Wind:

Parameter	Input Type	Range	Step	Default	Description
Turbine Diameter (m)	Number	20 – 200 m	10 m	200 m	Rotor diameter; grid cell Width = 3 × D, Height = 5 × D

Off-Shore Wind:

Parameter	Input Type	Range	Step	Default	Description
Turbine Diameter	Number	20 – 500 m	10 m	200 m	Rotor diameter; grid cell Width = 3 × D, Height = 5 ×

(m)					D
-----	--	--	--	--	---

For wind projects, the grid cell size is calculated automatically based on standard turbine spacing rules: the width is set to 3 times the turbine diameter, and the height to 5 times the diameter. For example, a 200m diameter turbine yields a 600m × 1000m grid cell. An information box below the input field displays the computed dimensions in real time.

Note: Input validation is applied in real time. If a value falls outside the allowed range, the input border turns red and an error message is displayed. The value is automatically clamped to the valid range when the field loses focus.

2.1.3 Map Preview

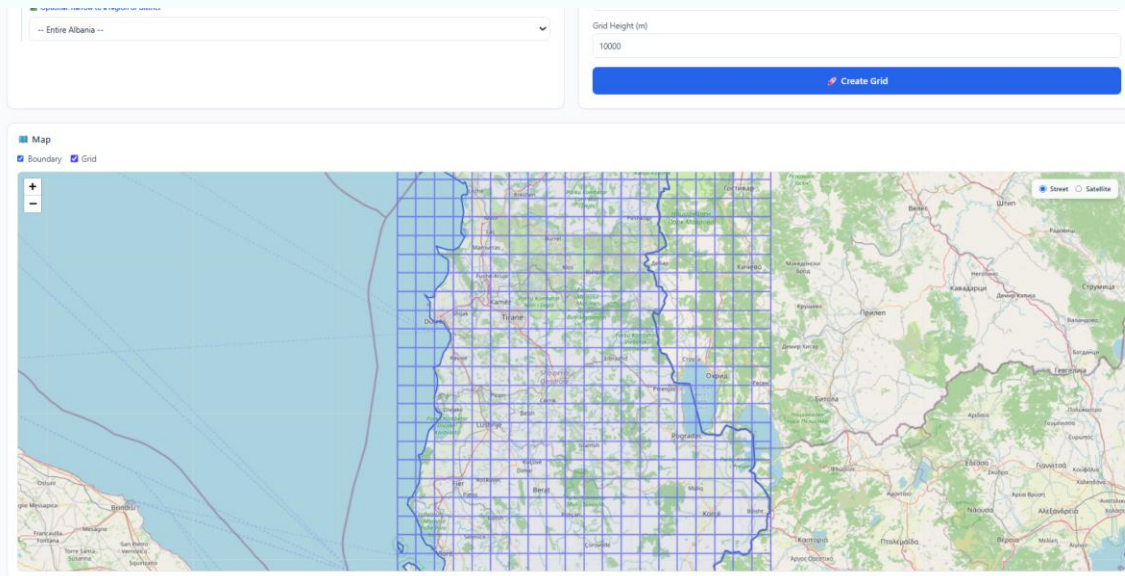


Figure 4: Map preview showing boundary polygon and grid overlay

When a boundary (country, EEZ zone, or Albania district) is selected, an interactive Leaflet map appears below the form. The map provides:

- **Boundary Polygon:** The selected region boundary displayed as a blue polygon with semi-transparent fill.
- **Grid Overlay:** A visualization of the grid cells as colored lines, computed in real time. If the projected cell count exceeds 300,000 cells, the grid overlay is suppressed with a warning message — the grid will still be created successfully but cannot be rendered at that density.
- **Base Map Toggle:** Two radio buttons in the top-right corner allow switching between "Street" (OpenStreetMap) and "Satellite" (ArcGIS World Imagery) base maps.
- **Layer Toggles:** Checkboxes to show/hide the boundary and grid overlay independently.
- **Grid Information:** A small info panel in the bottom-left corner displays the grid cell size and the estimated total number of cells.

2.1.4 Create Grid Button

The "🚀 Create Grid" button (blue, full-width) at the bottom of the parameters section initiates the grid generation process. When clicked:

5. A processing overlay appears with an animated spinner and the message "Creating grid cells..."
6. The backend FastGridEngine reprojects the boundary to Web Mercator (EPSG:3857), calculates the bounding box, and generates rectangular cells
7. Only cells whose centroids fall within the boundary polygon are retained
8. Each cell is assigned a unique cell_id and its geometry is stored as WKT (Well-Known Text)
9. The overlay disappears and results are displayed on the map and in the results panel

2.1.5 Upload Existing Grid

When the "Upload Existing Grid" mode is selected, a file drop area appears. The user can drag and drop a CSV file onto it, or click to open the file browser. The file is automatically uploaded upon selection via POST /api/grid/upload/. The CSV must contain at least the following columns:

- **cell_id** — a unique identifier for each grid cell
- **wkt** — the Well-Known Text geometry representation of each cell (EPSG:3857)

2.1.6 Grid Results

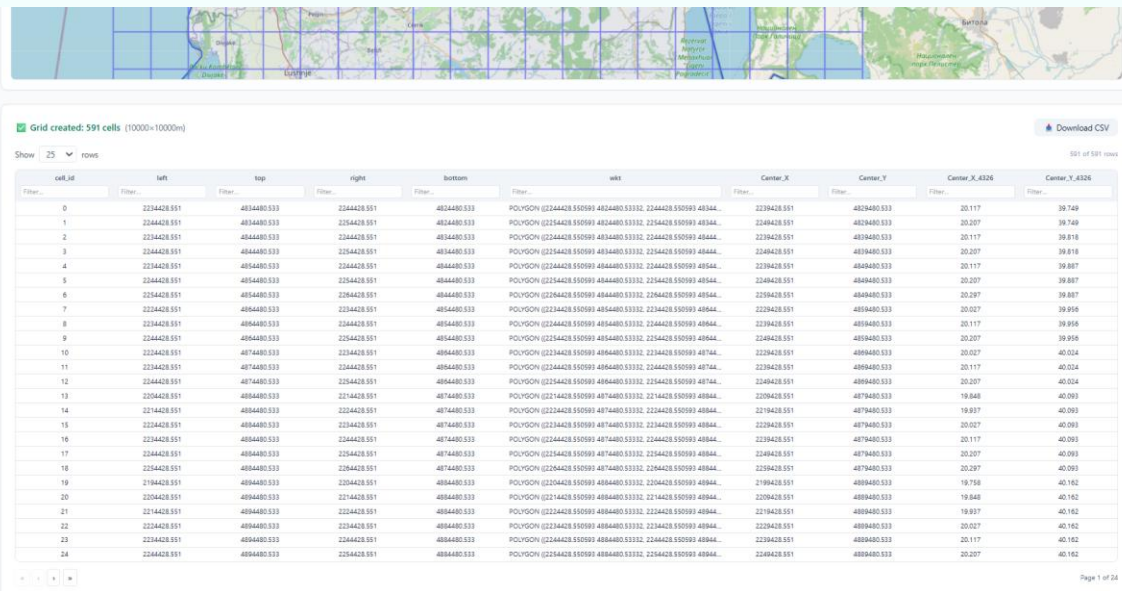


Figure 5: Grid results with success message and preview table

After successful grid creation, the following elements appear:

- **Success Message:** A green banner displays the confirmation message (e.g., "Grid created successfully") along with the total number of cells generated and the grid cell dimensions.
- **Download CSV Button:** Labeled "📄 Download CSV" — calls GET /api/grid/download/ and exports the full grid dataset as a comma-separated file (grid.csv, UTF-8 encoding).

- **Preview Table:** A scrollable AnalysisResultsTable showing the generated grid with pagination, sortable columns, and per-column text filters. Clicking a row focuses that cell on the map. Displays columns including cell_id, bounding coordinates (left, top, right, bottom), WKT geometry, and center point coordinates.
- **Map with Grid Cells:** The Leaflet map updates to display the actual generated grid cells aligned to real spatial coordinates.

2.2 Tab 2 — Layer Calculation

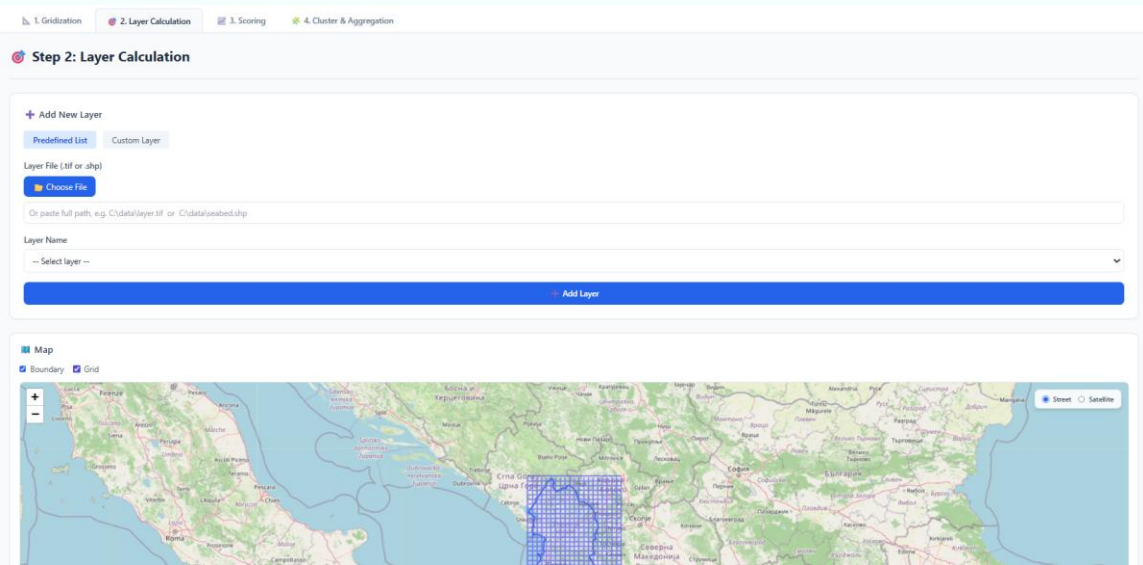


Figure 6: Layer Calculation tab with layers configured

The Layer Calculation tab is the second pipeline step. Here, the user configures which GIS raster (and optionally vector) layers to analyze, and how each layer should be processed for every grid cell. This step transforms raw spatial data into per-cell numerical metrics.

2.2.1 Prerequisites

This tab requires a grid to have been created in Step 1. If no grid data is available, a warning message is displayed: "No grid data available. Complete Step 1 first."

2.2.2 Add New Layer

The "Add New Layer" section provides a form to add spatial layers for analysis. Two modes are available via toggle buttons:

Predefined List Mode

This mode presents a dropdown of predefined layer names specific to the active project type. When a predefined layer is selected, the system automatically assigns the appropriate analysis modes based on the project configuration. Layers already added are removed from the dropdown.

For Solar PV, the predefined layers are organized into categories:

Category	Layer Names	Default Analysis Modes
Infrastructure – Transmission Lines	Distance to 110kV Line, Distance to 220kV Line, Distance to 400kV Line	distance
Infrastructure – Substations	Distance to 110kV Substation, Distance to 220kV Substation, Distance to 400kV Substation	distance

Land Use & Environment	Agricultural Areas, Forest, Urban/Residential/Industrial, Military, Protected Habitats	distance, coverage
Natural Resources	Energy Sources, Hydrography, Mineral Resources	distance, coverage
Risk & Climate	Natural Risk Zones, Slope (%), Solar Irradiation, Temperature	Varies: coverage, or min/max/mean
Transportation	Transport Networks	distance

Note: *On-Shore Wind and Off-Shore Wind modes have their own predefined layer lists. On-Shore adds Wind Speed (max/mean/min), Altitude (max), and Airports. Off-Shore uses Wind Speed, Bathymetry, Slope, Sea bed (shapefile), Ports, Subsea Cables, and other marine layers.*

Custom Layer Mode

In Custom mode, the user can define a new layer from scratch:

- **Custom Layer Name:** A free-text input for the layer name (e.g., "My Custom Layer").
- **Analysis Modes:** A set of toggle pill buttons for selecting one or more modes. Available modes are: distance, coverage, mean, max, min, median, std. Multiple modes can be selected simultaneously. Selected modes appear in blue.
- **Target Pixel Value:** Shown only when "distance" or "coverage" is among the selected modes. This numeric input (0–255) specifies which raster pixel value to target for distance calculation or coverage percentage computation. Default: 1.

Raster / Vector File Selection

Regardless of mode, a spatial data file must be specified for each layer. Two methods are available:

- **Native File Dialog / File Browser:** Click the "📁 Choose File" button. The system first attempts to open the OS native file picker (GET /api/native-file-dialog/). If the application is running headlessly (e.g., in Docker without a display), it falls back to the built-in FileBrowserModal. The browser shows the server directory structure, listing only .tif/.tiff and .shp files. Recent folders are stored in browser localStorage (up to 6 entries) for quick access.
- **Manual Path Entry:** A text field allows pasting the full file path manually (e.g., "C:\data\solar_irradiation.tif").

Shapefile (.shp) support: Shapefiles are accepted for Off-Shore Wind seabed substrate layers (EMODnet format). When a .shp file is selected, a note appears informing the user that it will be processed as a vector seabed layer using categorical substrate analysis.

Note: *All raster files must be in EPSG:3857 (Web Mercator) projection. The system validates the CRS on submission and rejects incompatible files. Maximum raster file size: 10 GB.*

Add Layer Button

Clicking the "+ Add Layer" button validates the input and adds the layer to the configured layers list via POST `/api/layers/add/`. Duplicate layer names are rejected with an error.

2.2.3 Configured Layers List

The screenshot shows a web interface titled "Step 2: Layer Calculation". It features a "Add New Layer" section with tabs for "Predefined List" and "Custom Layer". Below these is a "Layer File (.tif or .shp)" field with a "Choose File" button and a text input for pasting a full path. A "Layer Name" dropdown menu is also present. A large blue "Add Layer" button is at the bottom of this section. Below the form is a "Configured Layers (2)" section. It contains two layer cards: "Military Areas" (distance: Military_3857.tif) and "Hydrography" (distance: Hydro_3857.tif). Each card has a checkbox (checked for Military Areas, unchecked for Hydrography), an icon (tag for predefined, gear for custom), and a red "x" button. A green "Run Analysis" button is at the bottom of this section. At the bottom of the interface is a "Map" section with a map of Europe and a legend showing "Boundary", "Grid", "Military Areas", and "Hydrography".

Figure 7: Configured layers list with checkboxes and batch controls

Below the "Add New Layer" form, a "Configured Layers" section displays all currently added layers. The count is shown in the section title (e.g., "Configured Layers (5)").

Batch selection controls appear above the list:

- **All:** Checks all layer checkboxes — all layers will be included in the next analysis run.
- **None:** Unchecks all layer checkboxes — no layers will be included until manually checked.

Each layer card shows:

- A checkbox — checked layers are included in the analysis run; unchecked layers are skipped for the current run but remain configured
- An icon indicating predefined (tag icon) vs. custom (gear icon) origin
- The layer name
- The assigned analysis modes (comma-separated)
- The filename of the associated spatial file
- A red "x" button to permanently remove the layer (calls DELETE `/api/layers/<index>/remove/`)

Tip: Use checkboxes to selectively re-run analysis on specific layers without re-processing all layers. Only checked layers are passed to Run Analysis.

2.2.4 Layer Map Preview

An interactive Leaflet map (LayerMapPreview) below the configured layers list provides spatial context:

- **Boundary Layer:** The study area boundary polygon (toggleable).
- **Grid Layer:** The grid cells overlay (toggleable).
- **Raster Layer Previews:** Each configured raster layer can be toggled on/off individually. When toggled on for the first time, the system requests a server-rendered raster preview image (base64-encoded PNG with a blue-cyan-green-yellow-red colormap). The preview is overlaid on the map at 70% opacity.
- **Base Map Toggle:** Street (OpenStreetMap) or Satellite (ArcGIS) base maps.

2.2.5 Run Analysis

The "🚀 Run Analysis" button (green, full-width) starts the spatial analysis pipeline as an asynchronous task (POST /api/analysis/run-async/ with selected_indices). Only checked layers are included. The process:

10. A processing overlay appears with animated dots and a live progress message
11. The backend loads the grid data and reprojects it to match each raster CRS
12. For raster layers (.tif): the UniversalRasterScorer engine computes the requested metrics using memory-aware parallel chunking
13. For vector seabed layers (.shp): the seabed_scorer performs spatial intersection to determine the dominant substrate category per cell
14. Analysis modes are computed per cell: distance, coverage, mean, max, min, median, std, or categorical
15. Results are merged into a single DataFrame and stored in the session as scoring_results.pkl
16. The overlay disappears and results are displayed

Tip: Analysis time scales with cell count, layer count, and raster resolution. The system processes layers in parallel using multiple CPU workers. For large studies, use larger cell sizes to reduce processing time.

2.2.6 Analysis Modes Reference

Mode	Output Column Suffix	Description
distance	_dist_km	Distance (km) from each cell to the nearest pixel matching the target value, using GDAL proximity analysis with large-raster downsampling
coverage	_coverage_pct	Percentage (0–100) of the cell's area covered by pixels matching the target value
mean	_mean	Mean pixel value across all pixels within the cell boundary
max	_max	Maximum pixel value within the cell
min	_min	Minimum pixel value within the cell

median	_median	Median pixel value within the cell
std	_std	Standard deviation of pixel values within the cell
categorical	_dominant	Dominant substrate category (Off-Shore seabed .shp layers only)

2.2.7 Analysis Results

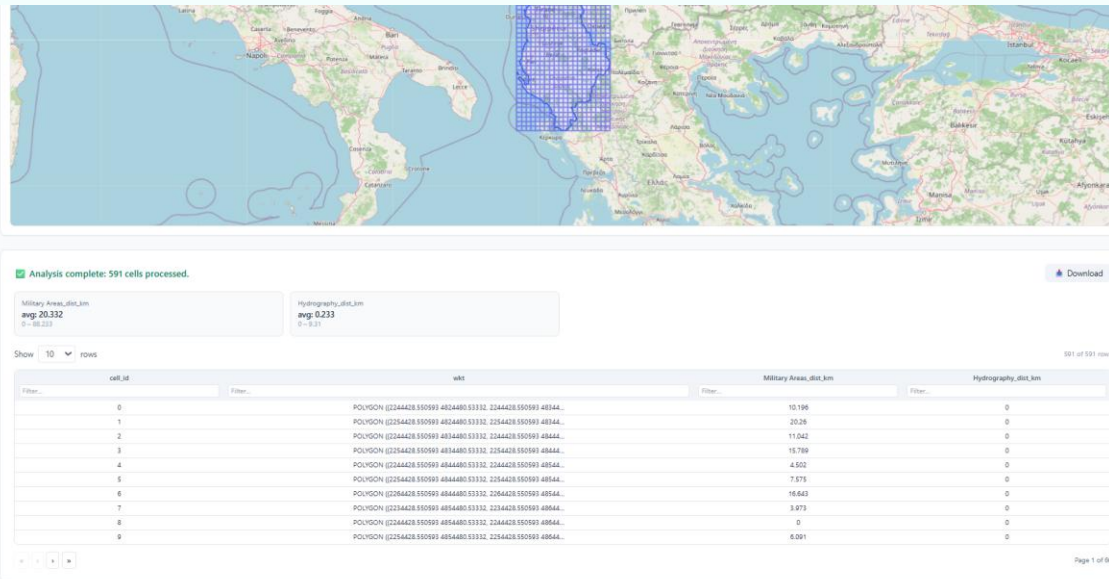


Figure 8: Analysis results with statistics cards and data table

After successful analysis, the results section displays:

Success Banner

A green-bordered container with the success message and a "Download" button for exporting the full results as raster_analysis_results.csv (semicolon-separated, comma decimals, UTF-8 BOM — compatible with Excel).

Statistics Cards

A grid of summary statistic cards (up to 4 per row) — one for each computed column. Each card shows the column name, the average value (avg), and the min–max range.

Data Table

A full-featured interactive AnalysisResultsTable (powered by TanStack React Table) displays the results:

- **Row Count Selector:** A dropdown to control rows per page (10, 25, 50, or 100).
- **Sortable Columns:** Click any column header to sort ascending (▲) or descending (▼).
- **Column Filters:** A text input below each column header enables real-time filtering.
- **Pagination:** Navigation buttons (first, previous, next, last) and a page indicator.
- **Number Formatting:** Numeric values are rounded to 3 decimal places.
- **Row Click:** Clicking any row focuses the corresponding grid cell on the map.

2.3 Tab 3 — Scoring

Figure 9: Scoring tab with layer configuration cards

The Scoring tab is the third pipeline step. Raw metric values from Step 2 are converted into normalized scores (0–100) using configurable scoring levels and weights. Hard exclusion constraints eliminate cells that violate critical thresholds. Transmission infrastructure layers (kV Lines and Substations) are handled via a dedicated "Connection" mode whose weights feed into cluster connection scoring in Step 4.

When the tab is opened, it automatically loads the available columns from the Step 2 analysis results and groups them by layer name based on column suffixes (`_dist_km`, `_coverage_pct`, `_mean`, `_max`, `_min`, `_median`, `_std`, `_dominant`). A configuration card is rendered for each detected layer group.

2.3.1 Prerequisites

This tab requires analysis results from Step 2. If no analysis data is available, a warning message is displayed: "No analysis data available. Complete Step 2 first."

2.3.2 Import CSV

A header button labeled "Import CSV" allows importing a previously saved scoring file. The uploaded CSV (via POST `/api/scoring/import-csv/`) must contain a `cell_id` column. This is useful to resume work from a previously exported scoring results file.

2.3.3 Weight Validation Banner

A persistent banner at the top of the configuration area displays the running total weight percentage across all active layers. The "🚀 Run Scoring" button is disabled until the total equals exactly 100%. The banner provides:

- Current total weight % (green when 100%, red otherwise)

- "Distribute Evenly" button — divides 100% equally across all layers in Scoring or Connection mode, overwriting current weights

2.3.4 Layer Configuration Cards

For each layer analyzed in Step 2, a configuration card is displayed. Four mode buttons in the card header control how the layer participates in scoring:

Mode	Description
Scoring	Layer metrics are scored using configurable level thresholds and contribute a weighted amount to FINAL_GRID_SCORE
Connection	Used for kV transmission layers (110/220/400 kV Lines and Substations). kV layers are auto-set to this mode. The configured weight is used in Step 4 Overall Score calculation, not in Step 3 FINAL_GRID_SCORE.
Exclusion	Hard knockout constraint — cells exceeding the threshold receive FINAL_GRID_SCORE = 0 permanently
Skip	Layer is excluded from all scoring and connection calculations

Scoring Mode — Layer Type: single_mode

Applies to layers with a single metric column (e.g., _mean, _max, or standalone _dist_km). Configuration options:

- **Weight (%):** Percentage contribution (0–100) of this layer to FINAL_GRID_SCORE.
- **Normalize by max:** Checkbox. When checked, all cell values are divided by the column maximum before applying level scoring, making scores relative to the study area's own range.
- **Scoring Levels (4 rows):** Level 1 (best) through Level 4 (worst). Each level has Min, Max, and Score fields. Adjacent level boundaries are linked: changing one level's Min auto-updates the adjacent level's Max. A level is highlighted red with a "min ≥ max" warning if invalid.

Scoring Mode — Layer Type: distance_coverage

Applies to layers with both _dist_km and _coverage_pct columns (e.g., Agricultural Areas, Forest). Configuration options:

- **Weight (%):** Percentage contribution to FINAL_GRID_SCORE.
- **Max Coverage Threshold (%):** Default 5%. If a cell's coverage percentage exceeds this threshold, the distance score is set to the minimum level (worst) regardless of the actual distance value. This penalizes cells that already significantly overlap with the obstacle. The constraint preview displays the formula (e.g., "Agricultural_Areas_coverage_pct ≤ 5%").
- **Distance Scoring Levels (4 rows):** Four levels mapping distance ranges to scores.

Scoring Mode — Layer Type: bathymetry_dual (Off-Shore Wind only)

Applies to depth-sensitive layers: Bathymetry, Wind Speed, and Slope in Off-Shore Wind mode. Accounts for two turbine foundation types depending on water depth:

- **Weight (%):** Percentage contribution to FINAL_GRID_SCORE.
- **Depth Threshold (m):** Default 60 m. Cells with depth $\leq$ threshold use the Bottom Fixed level table; cells deeper than the threshold use the Floating level table (for floating-foundation turbines).
- **Bottom Fixed Levels (4 rows):** Score levels applied within the bottom-fixed depth range.
- **Floating Levels (4 rows):** Score levels applied beyond the depth threshold.

Scoring Mode — Layer Type: seabed_categorical (Off-Shore Wind only)

Applies to EMODnet seabed substrate layers processed from shapefile data. Cells are scored based on the dominant substrate category:

EMODnet Substrate	Internal Category	Default Score
Sand, seabed	Sand	100
Coarse & mixed sediment	Gravel	70
Fine mud, muddy sand, sandy mud	Rack / Bad rack / Mud	40
Rock, boulder, Posidonia meadows	Boulder / Stony / Silt	0

Note: Seabed scoring only applies to bottom-fixed cells (depth $\leq$ depth threshold). Floating-depth cells receive score 0 from this layer type, as substrate is irrelevant for floating foundations.

Connection Mode Configuration

kV layer cards (110/220/400 kV Lines and Substations) are automatically set to Connection mode. Only one configuration field is shown:

- **Weight (%):** The percentage weight assigned to this connection asset type. Used in Step 4 cluster scoring (Overall Score calculation), not in Step 3 FINAL_GRID_SCORE.

Exclusion Mode Configuration

When a layer is set to "Exclusion" mode:

- **Metric Selector:** If the layer has multiple analysis columns, a dropdown selects which column to use as the constraint metric.
- **Threshold Value:** A numeric input for the exclusion limit.
- **Mode Selector:** Specifies whether the constraint is a maximum ($\leq$) or minimum ($\geq$) threshold.
- **Constraint Preview:** Displays the constraint formula (e.g., "Agricultural_Areas_coverage_pct $\leq$ 30") for clarity.

Warning: Cells violating an exclusion constraint have FINAL_GRID_SCORE forced to 0. These cells are excluded from cluster analysis in Step 4.

2.3.5 Run Scoring

The "🚀 Run Scoring" button (purple, full-width) is enabled only when total weights sum to 100%. Clicking it starts scoring as an asynchronous task (POST /api/scoring/run-async/). The process:

17. Collects scoring configuration (levels, weights, layer types) and exclusion constraints from all layer cards
18. For single_mode layers: maps metric values through 4 scoring levels to a 0–100 score
19. For distance_coverage layers: applies coverage threshold knockout, then distance levels
20. For bathymetry_dual layers: routes cells to bottom-fixed or floating level tables based on depth
21. For seabed_categorical layers: maps dominant substrate string to a configured score
22. Applies layer weight to compute each layer's weighted contribution
23. Sums all weighted contributions into FINAL_GRID_SCORE (0–100)
24. Applies hard exclusion constraints — violating cells set to FINAL_GRID_SCORE = 0
25. Records exclusion reasons (EXCLUSION_REASONS column) and stores results as final_scored_results.pkl

2.3.6 Scoring Results

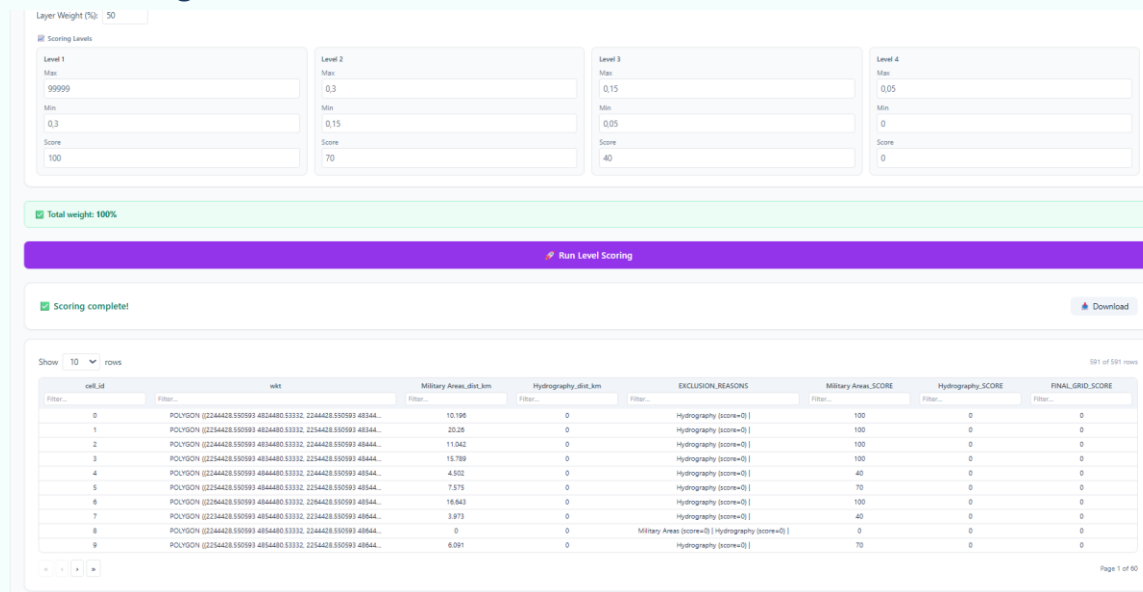


Figure 10: Scoring results with distribution cards and preview table

After scoring completes:

Score Distribution Cards

Category	Score Range	Color	Description
Excellent	≥ 80	Green	Highly suitable cells — top candidates
Good	60 – 79	Light Green	Well-suited cells with minor limitations

Fair	40 – 59	Yellow	Moderate suitability
Poor	20 – 39	Orange	Low suitability — significant constraints
Very Poor	1 – 19	Red	Unsuitable — multiple unfavorable factors
Excluded	= 0	Gray	Hard-excluded cells via constraint thresholds

Summary Statistics

- **Total:** Total number of cells processed
- **Excluded:** Cells with FINAL_GRID_SCORE = 0 due to hard constraints
- **Avg Score:** Mean FINAL_GRID_SCORE across non-excluded cells

Exclusion Summary Table

Lists each constraint layer with the column used, threshold applied, and number of cells excluded by that constraint.

Download & Data Preview

A "  Download" button exports the scored dataset via GET /api/scoring/download/ as final_scored_analysis.csv (semicolon-separated, comma decimals, UTF-8 BOM). An AnalysisResultsTable preview shows FINAL_GRID_SCORE, per-layer scores, and EXCLUSION_REASONS.

2.4 Tab 4 — Cluster & Aggregation

The screenshot displays the 'Step 4: Cluster Analysis' interface. At the top, a navigation bar includes four tabs: '1. Gridization', '2. Layer Calculation', '3. Scoring', and '4. Cluster & Aggregation'. Below this, the main heading is 'Step 4: Cluster Analysis'. Underneath, there are four sub-tabs: 'Connection Rules', 'Financial Constants' (which is active), 'Technical Constants', and 'Capacity Constraints'. The 'Financial Constants' sub-tab contains several input fields: '(S) PV CAPEX Per MW' with a value of 500000, '(S) Wind CAPEX Per MW' with a value of 1000000, 'Substation Installation Cost Ratio of CAPEX' with a value of 0.08, 'Line Expropriation Cost Ratio of CAPEX' with a value of 0.1, 'Land Expropriation Cost Ratio of CAPEX' with a value of 0.1, '(S) Transport Network Cost Per MW' with a value of 500, and '(S) Transport Network Base Cost' with a value of 400000. A 'Save Financial Constants' button is located at the bottom left of the sub-tab. At the bottom of the interface, there is a large blue button labeled 'Run Cluster Analysis & Scoring'.

Figure 11: Cluster tab with four configuration sub-tabs

The Cluster & Aggregation tab is the final pipeline step, combining spatial clustering, transmission connection scoring, and financial analysis. Adjacent high-scoring grid cells are grouped into project-sized clusters; each cluster is evaluated for its optimal grid connection and financial feasibility.

The tab is organized into four configuration sub-tabs (Capacity Constraints, Connection Rules, Financial Constants, Technical Constants) followed by the Run button.

2.4.1 Prerequisites

This tab requires scored results from Step 3. If no scoring data is available, a warning is displayed: "No scoring data available. Complete Step 3 first."

Alternatively, a pre-scored CSV file can be uploaded directly via POST `/api/cluster/upload-csv/` (must contain wkt and FINAL_GRID_SCORE columns; separator is auto-detected).

2.4.2 Sub-Tab A: Capacity Constraints (⚡)

1. Gridization
2. Layer Calculation
3. Scoring
4. Cluster & Aggregation

Step 4: Cluster Analysis

Connection Rules
Financial Constants
Technical Constants
Capacity Constraints

Nominal Capacity (MW)

Max Cluster Capacity (MW)

☒ Adjust capacity for coverage

Run Cluster Analysis & Scoring

Figure 12: Capacity Constraints sub-tab

Parameter	Input Type	Range / Step	Default	Description
Nominal Capacity (MW)	Number	Min 0.1, Step 0.5	13 MW	Assumed power capacity per grid cell. For wind, this represents the turbine nameplate rating.
Max Cluster Capacity (MW)	Number	Min 10, Step 10	250 MW	Maximum total capacity per cluster. Over-capacity connected components are split via greedy BFS to maintain spatial contiguity.
Adjust capacity for coverage	Checkbox	On / Off	Checked	Reduces each cell's effective capacity proportionally by its coverage percentages. A cell with 30% obstacle coverage gets 70% of nominal capacity.

2.4.3 Sub-Tab B: Connection Rules ()

1. Gridization
2. Layer Calculation
3. Scoring
4. Cluster & Aggregation

Step 4: Cluster Analysis

Connection Rules
Financial Constants
Technical Constants
Capacity Constraints

Criteria	Cap (MW)	Level 1			Level 2			Level 3			Level 4		
		Min	Max	Score	Min	Max	Score	Min	Max	Score	Min	Max	Score
110kV Line Line 110001	10-70	0.3	10	100	10	15	70	15	20	40	20	99999	10

Save Rules

Run Cluster Analysis & Scoring

Figure 13: Connection Rules editable table

An editable table of cluster connection scoring rules, pre-filtered to show only rules for kV layers configured in Step 2. If no kV layers were configured, a hint prompts the user to add kV layers in Step 2 to unlock connection scoring.

Each table row represents a scoring rule for a specific asset type and capacity range:

Column	Description
Criteria	Transmission asset name (read-only)
Cap MW Range	Cluster capacity range this rule applies to (0–30, 30–70, 70–180, or 180–400 MW)
L1 Min / Max / Score	Level 1 (closest/best): distance range (km) and assigned score
L2 Min / Max / Score	Level 2: distance range and score
L3 Min / Max / Score	Level 3: distance range and score
L4 Min / Max / Score	Level 4 (farthest/worst): distance range and score

The system evaluates each cluster's minimum distance to every configured transmission asset, applies the rules for the cluster's capacity range, and selects the best connection (highest score; ties broken by shortest distance). The "💾 Save Rules" button persists changes via PUT /api/scoring-rules/.

Note: For Off-Shore Wind mode, only 220kV and 400kV assets are available. The rules table adjusts to show only offshore-relevant entries.

2.4.4 Sub-Tab C: Financial Constants (💰)

1. Gridization2. Layer Calculation3. Scoring4. Cluster & Aggregation

Step 4: Cluster Analysis

Connection RulesFinancial ConstantsTechnical ConstantsCapacity Constraints

(S) PV CAPEX Per MW
500000

Line Depreciation Cost Ratio of CAPEX
0.1

(S) Transport Network Cost Per MW
500

(S) Wind CAPEX Per MW
1000000

Land Depreciation Cost Ratio of CAPEX
0.1

Substation Installation Cost Ratio of CAPEX
0.08

(S) Transport Network Base Cost
400000

Save Financial Constants

Run Cluster Analysis & Scoring

Figure 14: Financial Constants editing grid

Editable financial parameters used for CAPEX and LCOE calculations. Fields are shown or hidden based on the active project mode.

Parameter	Default	Applies To	Description
PV CAPEX per MW	\$500,000	Solar PV	Capital cost per MW for PV installations
Wind CAPEX per MW	\$1,000,000	Wind	Capital cost per MW for wind installations
Substation Installation	Solar: 8%,	All	Substation cost as fraction of generation

Cost Ratio	Wind: 6%		CAPEX
Line Expropriation Cost Ratio	10%	All	Land expropriation cost for line corridor
Land Expropriation Cost Ratio	10%	All	Land acquisition as fraction of CAPEX
Transport Network Base Cost	\$400,000	Wind	Fixed base cost for access infrastructure
Transport Network Cost per MW	\$500 / MW	Wind	Variable transport cost per installed MW

Transmission cost table (from financial_constants.json):

Type	kV	Capacity Range (MW)	Cost/km	Fixed Cost
Line	110	0 – 30	\$170,000	\$0
Line	110	30 – 70	\$170,000	\$0
Line	220	70 – 180	\$280,000	\$0
Line	400	180 – 400	\$400,000	\$0
Substation	110	0 – 30	\$170,000	\$500,000
Substation	110	30 – 70	\$170,000	\$1,000,000
Substation	220	70 – 180	\$280,000	\$3,000,000
Substation	400	180 – 400	\$400,000	\$8,000,000

The "💾 Save Financial Constants" button persists changes via PUT /api/financial-constants/.

2.4.5 Sub-Tab D: Technical Constants (⚙️)

The screenshot displays the 'Step 4: Cluster Analysis' sub-tab, with the 'Technical Constants' section active. It features a 'Capacity Factor' input field with a placeholder text 'Leave empty for auto-calculation'. Below the input field is a 'Save Technical Constants' button. At the bottom of the sub-tab, there is a blue button labeled 'Run Cluster Analysis & Scoring'.

Figure 15: Technical Constants sub-tab

- **Capacity Factor Override (0–1):** Optional numeric input. If set, overrides the per-cluster calculated capacity factor for all clusters. If left empty, each cluster's CF is calculated from its energy yield and installed capacity.

- **Cp Values Table (Wind projects only):** Editable scrollable table of wind speed (m/s) vs. power coefficient (Cp) pairs. Defines the turbine power curve. Matches the specific turbine model's aerodynamic performance. The "💾 Save Technical Constants" button persists changes via PUT /api/cp-values/.

Tip: Populate the Cp table with the manufacturer's power curve for the turbine model under consideration. This significantly affects energy yield and LCOE accuracy.

2.4.6 Run Cluster Analysis

The "🚀 Run Cluster Analysis" button (indigo, full-width) initiates the complete pipeline as an asynchronous task (POST /api/cluster/run-async/). Three phases:

Phase 1: Spatial Clustering (ClusterEngine)

26. Loads scored data, filters out cells with FINAL_GRID_SCORE ≤ 0
27. Calculates effective capacity per cell (accounting for clipped edge cells and optional coverage reduction)
28. Performs spatial adjacency analysis — identifies cells sharing a boundary via spatial intersection join
29. Builds a NetworkX graph with cells as nodes and shared edges as graph edges
30. Identifies connected components (candidate clusters)
31. Enforces max cluster capacity: over-capacity components split via greedy BFS maintaining spatial contiguity
32. Dissolves cell geometries into a unified cluster polygon per cluster
33. Aggregates metrics: sum for capacity, mean for scores/coverages, min for all distance columns

Phase 2: Connection Scoring (ClusterScorer)

34. For each cluster, finds minimum distance to each configured transmission asset type across all member cells
35. Evaluates each asset against connection rules for the cluster's capacity range
36. Selects optimal connection: highest score; ties broken by shortest distance
37. Records winning connection type, kV level, distance, and score
38. Wind only: classifies cluster as "Distribution" (< 30 MW) or "Transmission" (≥ 30 MW) grid
39. Computes Overall Score = Mean Cell FINAL_GRID_SCORE + (connection weight $\times$ connection score)

Phase 3: Financial Analysis (FinancialScorer)

Solar PV Financial Model:

- CAPEX OF PV = Capacity $\times$ PV CAPEX per MW
- SUBSTATION COST = CAPEX OF PV $\times$ Substation PV Ratio (8%)
- LAND COST = CAPEX OF PV $\times$ Land Cost Ratio (10%)
- SLOPE COST = (CAPEX OF PV $\times$ Mean Slope % $\times 9/15$) / 100
- LINE CAPEX = Connection Distance $\times$ Cost per km (+ Fixed Cost for substation connections)

- $\text{LINE EXPROPRIATION} = \text{LINE CAPEX} \times \text{Line Expropriation Ratio (10\%)}$
- $\text{TOTAL CAPEX} = \text{PV} + \text{SUBSTATION} + \text{LAND} + \text{SLOPE} + \text{LINE} + \text{LINE_EXPROPRIATION}$
- $\text{Yearly Energy (MWh)} = 1688 \times \text{Solar Irradiation Fraction} \times \text{Capacity} \pm \text{Temperature Correction}$
- $\text{Capacity Factor} = \text{Yearly Energy} / (8,760 \times \text{Capacity})$
- $\text{LCOE (\$/MWh)} = (\text{Annualized CAPEX} + \text{Annual OPEX}) / \text{Yearly Energy}$ [25-yr lifetime, 8% discount, 2% OPEX]
- $\text{Payback Period (Yrs)} = \text{TOTAL CAPEX} / (\text{Yearly Energy} \times \$50/\text{MWh})$

Wind Financial Model (On-Shore & Off-Shore):

- $\text{CAPEX} = \text{Capacity} \times \text{Wind CAPEX per MW}$
- $\text{SUBSTATION COST} = \text{CAPEX} \times \text{Substation Wind Ratio (6\%)}$
- $\text{LAND COST} = \text{CAPEX} \times \text{Land Cost Ratio (10\%)}$
- $\text{TRANSPORT NETWORKS} = (\text{Transport Distance} \times \$400,000) + (\text{MW} / 4 \times \$2,000)$
- $\text{LINE CAPEX} = \text{Connection Distance} \times \text{Cost per km (+ Fixed Cost for substation connections)}$
- $\text{LINE EXPROPRIATION} = \text{LINE CAPEX} \times \text{Line Expropriation Ratio (10\%)}$
- $\text{TOTAL CAPEX} = \text{CAPEX} + \text{SUBSTATION} + \text{LAND} + \text{TRANSPORT} + \text{LINE} + \text{LINE_EXPROPRIATION}$
- $\text{Air Density } (\rho) = 1.225 - 0.264 \times (\text{Altitude} / 2000)$ [elevation-based density correction]
- $C_p = \text{nearest-neighbor lookup from } C_p \text{ values table (wind speed} \rightarrow \text{power coefficient)}$
- $\text{Swept Area} = \pi \times \text{rotor_radius}^2$
- $\text{Yearly Energy (MWh)} = (0.5 \times \rho \times \text{Area} \times v^3 \times C_p \times 8,760 / 1,000,000) \times (\text{MW} / 4.2)$
- LCOE (\$/MWh) and Payback Period calculated using same assumptions as Solar

Additional Output Metrics:

- $\text{CAPEX/MW (\$)} = \text{TOTAL CAPEX} / \text{Installed Capacity}$
- $\text{Scaled Overall Score} = \text{Overall Score} \times (\text{Capacity Factor} / \text{Max CF across all clusters})$

2.4.7 Cluster Results

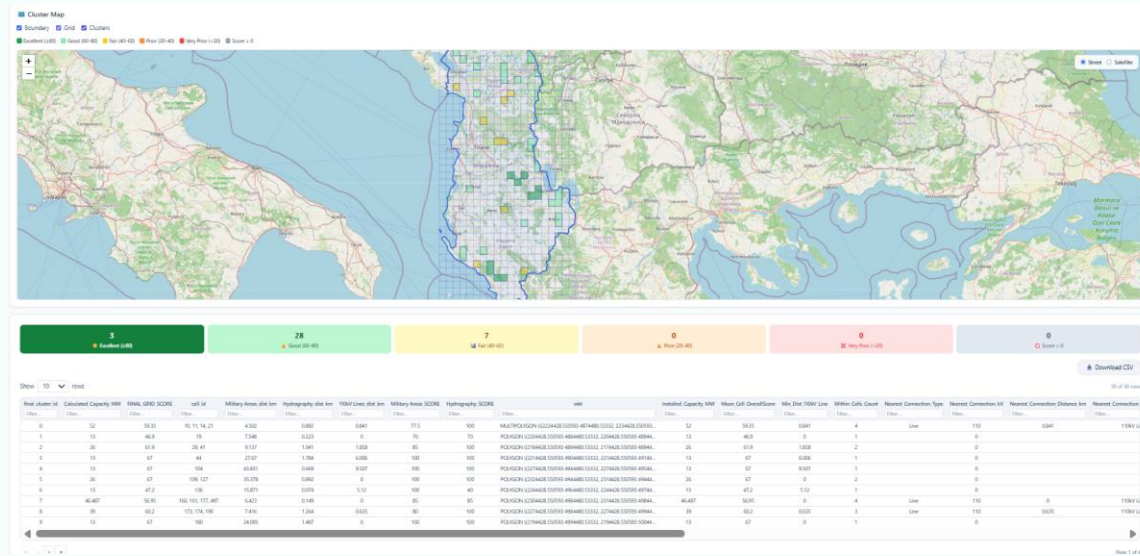


Figure 16: Cluster results with statistics, map, and data table

Success Banner & Download

A green banner with a "Download" button exports clustered_scored_results.csv (semicolon-separated, comma decimals, UTF-8 BOM; geometry column excluded, wkt preserved).

Summary Statistics Cards

Metric	Description
Total Clusters	Total number of clusters formed from eligible cells
Avg Capacity (MW)	Average installed capacity per cluster
Total Capacity (MW)	Sum of all cluster capacities
Avg Overall Score	Mean Overall Score across all clusters
Avg LCOE (\$/MWh)	Average Levelized Cost of Energy

Connection Distribution Cards

- Connection Type Distribution:** Count and percentage of clusters grouped by winning connection type (Line vs. Substation).
- kV Distribution:** Count and percentage of clusters grouped by voltage level (110kV, 220kV, 400kV).

Cluster Map (ClusterMapPreview)

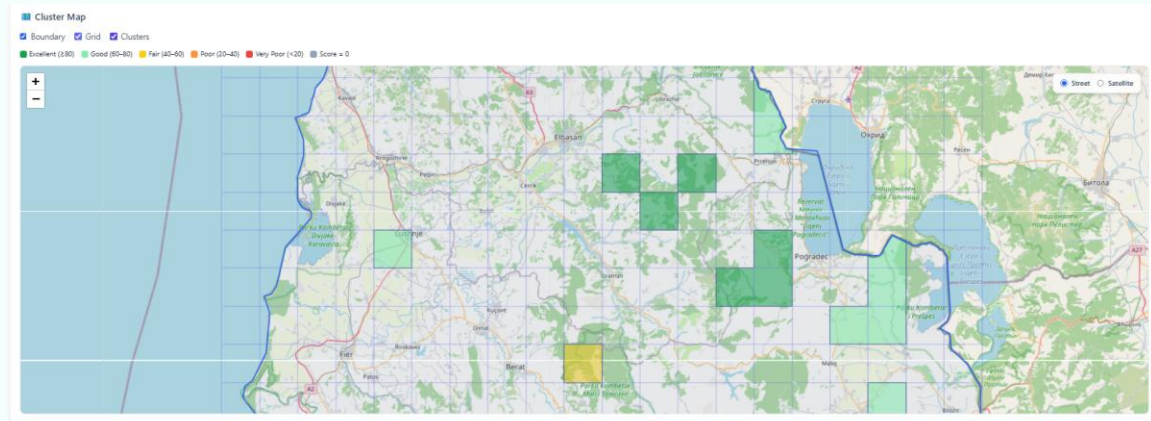


Figure 17: Cluster map with colored polygon overlays

An interactive Leaflet map displays all cluster polygons as colored overlays. Each polygon represents the dissolved geometry of all member cells. Clicking a cluster polygon highlights it and scrolls the data table to that row. Layer toggles and base map selection are available.

Cluster Data Table

An AnalysisResultsTable shows per-cluster details: cluster ID, installed capacity (MW), number of member cells (Within_Cells_Count), best connection type/kV/distance/score, overall score, total CAPEX, CAPEX/MW, yearly energy (MWh), LCOE (\$/MWh), capacity factor, payback period (years), and component cost breakdowns. Supports sorting, filtering, pagination, and row-click map focus.

3. Project Mode Differences

While the four-tab pipeline structure is identical across all project modes, each mode has distinct parameters, layers, and scoring configurations. This section summarizes the key differences.

3.1 Grid Parameters

Parameter	Solar PV	On-Shore Wind	Off-Shore Wind
Grid Input	Width × Height (m)	Turbine Diameter → 3D × 5D	Turbine Diameter → 3D × 5D
Turbine Diameter Range	N/A	20 – 200 m, step 10	20 – 500 m, step 10
Default Cell Size	1,000 m × 1,000 m	600 m × 1,000 m (200 m turbine)	600 m × 1,000 m (200 m turbine)
Boundary Source	Country (NUTS) or Albania ADM	Country (NUTS) or Albania ADM	EEZ (Maritime Zones)

3.2 Predefined Layers

Layer Category	Solar PV	On-Shore Wind	Off-Shore Wind
Wind Resources	—	Wind Speed (max, mean, min)	Wind Speed (max, mean, min) — bathymetry_dual
Solar Resources	Solar Irradiation (mean), Temperature (mean)	—	—
Terrain	Slope (%) min/max/mean	Slope (%), Altitude (max)	Slope (%) bathymetry_dual; Bathymetry (max) bathymetry_dual
Seabed	—	—	Sea bed (.shp EMODnet) seabed_categorical
Grid Infrastructure	110/220/400 kV Lines & Substations (distance)	110/220/400 kV Lines & Substations (distance)	220/400 kV Lines & Substations only (distance)
Land/Marine Use	Agriculture, Forest, Urban, Military, Protected (dist+cov)	Agriculture, Airports, Forest, Land Use, Military, Protected (dist+cov)	Fishing, Military, Shipping, Tourism, Protected (dist+cov)
Transportation	Transport Networks (distance)	Transport Networks (distance)	Ports (distance), Subsea Cables (distance)
Natural Resources	Energy Sources, Hydrography, Minerals	Energy, Hydrography, Minerals, Natural Risk	Natural Risk

3.3 Scoring Types Used per Mode

Scoring Type	Solar PV	On-Shore Wind	Off-Shore Wind
single_mode	Yes — Slope, Irradiation, Temperature	Yes — Wind, Altitude, Slope	Yes — some marine layers
distance_coverage	Yes — Agriculture, Forest, Urban, etc.	Yes — Agriculture, Airport, Forest, etc.	Yes — Fishing, Military, Shipping, etc.
bathymetry_dual	No	No	Yes — Bathymetry, Wind Speed, Slope
seabed_categorical	No	No	Yes — Sea bed substrate

3.4 Color Theme

Element	Solar PV	On-Shore Wind	Off-Shore Wind
Header Accent	Orange	Dark Blue (Navy)	Cyan / Blue
Card Border	Orange-300	Blue-400	Cyan-300
Selection Button	Orange	Dark Blue	Blue

3.5 Financial Model Differences

Parameter	Solar PV	Wind (On-Shore & Off-Shore)
Base CAPEX	\$500,000 / MW	\$1,000,000 / MW
Substation Ratio	8% of CAPEX	6% of CAPEX
Slope Cost	Included (terrain correction)	Not applicable
Transport Cost	Not applicable	Included (access road infrastructure)
Energy Calculation	$1688 \times \text{irradiance fraction} \times \text{capacity} \pm \text{temperature correction}$	$0.5 \times \rho \times \text{swept area} \times v^3 \times C_p \times 8,760 \times (\text{MW}/4.2)$
Air Density Correction	No	Yes — $\rho = 1.225 - 0.264 \times (\text{altitude}/2000)$
Capacity Factor Source	Solar irradiation-based	Wind speed & Cp table-based

4. Appendices

4.1 API Endpoints Reference

All API endpoints are served at <http://localhost:8000/api/>. All requests requiring session context must include the X-Session-ID header (UUID stored in browser localStorage as dashboard_session_id).

Category	Method	Endpoint	Description
Project	POST	/project/select/	Create session: project_type = Solar OnShore OffShore
Project	GET	/project/status/	Get session state and step completion flags
Project	POST	/project/reset/	Clear session; keep_project_type: bool
Project	GET	/project/config/	Get layer/scoring config for active project type
Grid	POST	/grid/create/	Create grid from country/EEZ/Albania boundary
Grid	POST	/grid/upload/	Upload existing grid CSV
Grid	GET	/grid/download/	Download grid.csv
Grid	GET	/countries/	List NUTS European countries
Grid	GET	/eez-zones/	List EEZ maritime zones
Grid	GET	/albania/regions/	List Albania ADM1 regions
Grid	GET	/albania/districts/	List Albania ADM2 districts (filter: ?region=...)
Layers	POST	/layers/add/	Add raster (.tif) or vector (.shp) layer
Layers	DELETE	/layers/<index>/remove/	Remove layer by index
Analysis	POST	/analysis/run-async/	Start async raster/vector analysis (returns task_id)
Analysis	GET	/analysis/download/	Download raster_analysis_results.csv
Scoring	POST	/scoring/run-async/	Start async level scoring (returns task_id)
Scoring	GET	/scoring/download/	Download final_scored_analysis.csv
Scoring	POST	/scoring/import-csv/	Import pre-computed scoring CSV
Cluster	POST	/cluster/run-async/	Start async cluster analysis (returns task_id)
Cluster	GET	/cluster/download/	Download clustered_scored_results.csv
Cluster	POST	/cluster/upload-csv/	Upload pre-scored CSV as cluster input
Config	GET/PUT	/financial-constants/	Read / write financial parameters

Config	GET/PUT	/cp-values/	Read / write wind turbine Cp table
Config	GET/PUT	/scoring-rules/	Read / write cluster connection scoring rules
Tasks	GET	/task/<task_id>/progress/	Poll async task progress: {progress, message, status}
Browse	GET	/browse/	Browse server filesystem for .tif / .shp files
Browse	GET	/native-file-dialog/	Open OS native file picker dialog

4.2 Glossary of Terms

Term	Definition
CAPEX	Capital Expenditure — total upfront investment cost
Cp (Power Coefficient)	Fraction of wind energy extracted by a turbine (0 to Betz limit ~0.59). Defined via lookup table.
CRS / EPSG	Coordinate Reference System / European Petroleum Survey Group code. EPSG:3857 (Web Mercator) required for all input rasters.
EEZ	Exclusive Economic Zone — maritime area (200 NM) with coastal state resource rights. Used as boundary for Off-Shore Wind mode.
EMODnet	European Marine Observation and Data Network — provider of seabed substrate shapefile data.
GeoJSON	Open standard format for encoding geographic data structures (geometries + attributes).
Grid Cell	Single rectangular polygon unit identified by a unique cell_id.
LCOE	Levelized Cost of Energy — total lifetime cost per MWh produced.
NUTS	Nomenclature of Territorial Units for Statistics — EU administrative division standard.
Raster	Pixel-based spatial data format (GeoTIFF) for continuous variables.
Session ID	UUID per project session, stored in localStorage and sent in X-Session-ID header.
Task ID	UUID returned by async endpoints; used to poll /api/task/<id>/progress/.
WKT	Well-Known Text — text representation of vector geometries.
Shapefile	Geospatial vector format (.shp + companion files). Supported for seabed substrate layers.

4.3 Data Requirements

Input Raster Files (.tif / .tiff)

- Format: GeoTIFF (.tif / .tiff)
- Coordinate Reference System: EPSG:3857 (Web Mercator) — mandatory

- Maximum file size: 10 GB
- Recommended resolution: match or finer than the intended grid cell size
- Storage: place files in the data/ directory or specify the full server-side path during layer configuration

Input Vector Files (.shp — Off-Shore Seabed only)

- Format: ESRI Shapefile (.shp with companion .dbf, .shx, .prj files)
- Data source: EMODnet seabed substrate classification
- Must contain substrate classification attributes compatible with EMODnet schema
- The seabed scorer performs spatial intersection with grid cells and assigns dominant substrate category

Upload Grid CSV Columns

Column	Type	Required	Description
cell_id	Integer	Yes	Unique identifier for each cell
wkt	String (WKT)	Yes	Cell geometry in Well-Known Text format (EPSG:3857)

Download Output File Formats

Download	Filename	Separator	Decimal	Encoding
Grid	grid.csv	Comma (,)	Dot (.)	UTF-8
Raster Analysis	raster_analysis_results.csv	Semicolon (;)	Comma (,)	UTF-8 BOM
Scored Results	final_scored_analysis.csv	Semicolon (;)	Comma (,)	UTF-8 BOM
Cluster Results	clustered_scored_results.csv	Semicolon (;)	Comma (,)	UTF-8 BOM

Note: UTF-8 BOM encoding ensures correct display of special characters when opening CSV files in Microsoft Excel on Windows.

4.4 Tips and Best Practices

- Use the file browser's "Recent" button to quickly navigate to the last 6 visited directories
- The file browser persists the last directory across sessions via POST /api/browse/save-last-dir/
- Column headers in all data tables are clickable for sorting — click once ascending, again descending
- Use column filter inputs to search for specific cells or clusters by any attribute value
- The map supports standard Leaflet controls: scroll wheel to zoom, click-drag to pan
- Switch Street/Satellite base maps with the radio buttons in the map's top-right corner
- Download results at each step as a checkpoint — protects work against Docker restart or session loss

- Use "Distribute Evenly" in the Scoring tab to quickly set equal weights, then fine-tune individually
- Use layer checkboxes in Step 2 to selectively re-run analysis on specific layers only
- Scoring level boundaries are linked — changing one level's Min auto-updates the adjacent Max
- Review the Score Distribution cards after Step 3 — if too many cells are Excluded, loosen exclusion thresholds

4.5 Troubleshooting

Issue	Possible Cause	Solution
"No grid data available" warning	Step 1 not completed	Go to Gridization tab and create or upload a grid
Raster file rejected with CRS error	File is not in EPSG:3857	Reproject using QGIS or gdalwarp before adding it
Run Analysis button disabled	No layers checked for inclusion	Use the All link or check individual layer checkboxes
Run Scoring button disabled	Total weights $\neq$ 100%	Use "Distribute Evenly" or adjust weights manually until total = 100%
Analysis runs very slowly	Large study area or high-resolution rasters	Use larger grid cells or lower-resolution rasters to reduce processing time
Session data lost after restart	Docker container was restarted	Download CSV at each step to preserve work; re-upload in Step 4 if needed
Map preview not loading	Browser rendering issue	Refresh the page (Ctrl+Shift+R) or clear browser cache
File browser shows empty directory	Wrong path or Docker volume not mounted	Ensure data/ directory is mounted in Docker Compose and contains .tif / .shp files
Albania sub-district dropdown not visible	Region not selected yet	Select a Region first; the District dropdown appears after region selection
Shapefile returns empty seabed results	Missing companion files or CRS mismatch	Ensure .shp, .dbf, .shx files are all present; reproject if needed
Cluster results: 0 clusters formed	All cells excluded (score = 0)	Loosen exclusion constraints in Step 3; thresholds may be too strict
Connection Rules table is empty	No kV layers in Step 2	Add at least one kV transmission layer in Step 2 to enable connection scoring
Scoring weights won't reach 100%	Rounding in manual entry	Use "Distribute Evenly", then fine-tune individual layer weights